

Ruyan Qin

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MSc Virtual and Augmented Reality candidate with hands-on Unity, C#, Meta Quest and XR Interaction Toolkit experience. Independently develops XR gameplay, gesture and physics-based interaction, UI feedback and working prototypes, combining implementation with user research and behavioral data. Increased a WebGL game's average playtime from 44 seconds to 1 minute 53 seconds through three Poki test rounds of approximately 500 players each. Expected to complete the MSc in September 2026 and available from October 2026.

Education

- **Goldsmiths, University of London** **London, United Kingdom**
MSc Virtual and Augmented Reality – in progress *2025.09-2026.09 expected*
Game programming, augmented reality, virtual reality, game-method design and motion capture.
- **University of Glasgow** **Glasgow, United Kingdom**
BSc Computer Science *2022.09-2025.07*
Database systems, software engineering, web development, algorithms and data structures, HCI and extended reality development.

Core Competencies

- **Unity and C#:** gameplay systems, UI, animation, particles, WebGL builds, Git workflows and reusable package integration.
- **XR Development:** XR Interaction Toolkit, Meta XR SDK, Auto Hand, WebXR, Meta Quest 3, gesture-driven and physics-based interaction.
- **Hardware and Unity Communication:** Arduino-to-Unity network communication.
- **Interaction Design:** gameplay and interaction flows, rapid prototyping, UI and multimodal feedback, physical/digital experience design.
- **User Research:** questionnaires, interviews, usability testing, behavioral-data analysis and evidence-led iteration.

Professional Experience

- **BrainCo (Shenzhen Branch)** **Shenzhen, China – onsite**
Unity Real-time Interaction Development Intern *2026.07-2026.08*
 - **Real-time prototypes:** Designed and developed three Unity game prototypes for EMG wristband, IMU glove and gesture-recognition scenarios, covering gameplay, input integration and prototype testing.
 - **Reusable input package:** Built a Unity package that maps external hardware events to virtual Input System devices, supporting multiple configurations, automatic reconnection, event deduplication and keyboard/mouse simulation.
 - **Gesture-based gameplay:** Integrated real-time gesture signals delivered via SSE into wall climbing, dashing, collision and projectile mechanics, with character animation and particle feedback.
 - **IMU glove interaction:** Used wrist angle, acceleration and finger-bend data to implement two-handed control and dynamic grab evaluation in a 3D claw-machine game.
 - **EMG interaction:** Integrated wristband signals delivered via WebSocket into movement, dragging, object pickup, chopping and directional throwing mechanics.
- **Dream Reality Interactive** **United Kingdom – remote, part-time**
Unity Game Development and Design Intern *2026.05-2026.07*
 - **Independent prototype:** Developed the tutorial and first level of the Unity WebGL game *Pizza Delivery*, covering gameplay, interaction, level flow and release.
 - **Data-led iteration:** Ran three Poki test rounds of approximately 500 players each, increasing average playtime from 44 seconds to 1 minute 53 seconds (approximately 157%).
 - **Core loop design:** Created standard, timed and VIP orders with rewards, countdowns and area unlocks, forming an order-delivery-upgrade progression loop.
 - **Experience and delivery:** Added audio, particles, camera and UI feedback, refined onboarding, managed versions with Git and configured automated WebGL builds for Poki and itch.io.

- **Liquid City – Plant Bot** **United Kingdom – onsite**
Interaction Design and Arduino Development Intern *2026.05*
 - **End-to-end prototype:** Contributed to Plant Bot from user research, concept design and feature definition through physical-prototype validation during a one-month internship.
 - **Data-to-emotion design:** Mapped soil moisture, light, temperature and humidity to six emotional states and care prompts, and designed the web UI and information architecture.
 - **Physical/digital integration:** Supported Arduino/ESP32 sensor integration, circuit and PCB work, soldering, enclosure modeling, 3D printing and assembly; connected real-time data to a Unity AR scene.
- **Suzhou Dejie Wei Digital Technology Co., Ltd.** *2025.12-2026.01*
VR Developer
 - **VR interaction:** Developed gameplay and interaction features and refined locomotion and drawing-control logic to reduce unintended input and improve controllability.
 - **Testing and communication:** Conducted user testing and implemented Unity client-server data exchange for dynamic content updates and real-time state feedback.
- **University of Glasgow – AugSoc Research Project** **Glasgow, United Kingdom**
VR Developer and User Researcher *2025.06-2025.09*
 - **AR creation tool:** Built a Meta Quest 3 prototype with 2D/3D modes, spray painting, path drawing, dragging and scaling.
 - **User research:** Conducted an 18-participant questionnaire and interview study and translated findings into interaction and feedback improvements.
- **University of Glasgow** **Glasgow, United Kingdom**
Computer Laboratory Teaching Assistant *2024.09-2025.03*
 - **Technical instruction:** Supported students with problem-solving, practical laboratory skills, group tutorials and project implementation.

Selected Projects

- **Abyss – Immersive locomotion and storytelling (2026):** Mapped dynamic hand movement to locomotion and combined large-scale fear environments, spatial audio and event-triggered narrative in VR.
- **Emotional Mask – Rapid gameplay delivery (2026 Game Jam):** Contributed gameplay, level logic and narrative flow and implemented level-linked gesture triggers. github.com/kjie11/EmotionalMask
- **AR Escape Room – Project delivery (2025):** Designed physical/digital puzzles using virtual-window erasure, handwriting and QR triggers; led task planning, code integration and debugging. github.com/SophiaSun0819/TheUncagedPact
- **Speaking World – Multimodal AI interaction (2025):** Combined object recognition, AI dialogue, voice feedback and AR surface drawing into a real-time interaction flow.
- **WebXR Interactive Gallery – Mobile immersive experience (2024):** Implemented rotation, gaze, shake, voice and gesture interaction and iterated feedback through usability testing.